How Much Does the Batting Order Matter?

Exact Expected-Runs Optimization with an Empirical-Bayes Plate-Appearance Model, and the 2026 World Series

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November 2, 2026

Abstract

We study the choice of a batting order for a fixed set of nine hitters, with the objective of maximizing the expected number of runs scored in nine offensive innings. Each hitter's plate-appearance outcome distribution against the opposing starting pitcher and bullpen is estimated, using only information available before the game, by an empirical-Bayes multiplicative (log5-type) model fitted to Statcast data. Base-out transitions, including base-running between plate appearances, are estimated from the data rather than assumed. Under explicit Markov assumptions the expected runs of a lineup can be computed exactly through an inning-level Markov chain on the leadoff slot; we prove that the computation is well defined, bound its truncation error, and use rotational structure to evaluate all 9! = 362,880 orders in about 1.40 seconds, so that the reported optimum is certified within the model. On held-out seasons (2025–2026) the plate-appearance model improves log-loss over the league baseline by 0.0272 nats per plate appearance, and the run predictions are calibrated (slope 0.94). Over 9,860 team-games, the model-optimal order exceeds the order actually used by 0.020 expected runs per game (3.2 per 162 games), and actual orders rank, on median, above 97.8% of all possible orders. For Game 5 of the 2026 World Series (Milwaukee 4, Tampa Bay 1) the estimated gains are +0.010 (Milwaukee) and +0.024 (Tampa Bay) runs per game, with 90% parameter- uncertainty intervals that are of the same order as the gains themselves. These are model-based counterfactuals: a historical game reveals only the outcome of the order that was used, so no observed result can confirm them.

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*Code, data-processing scripts and the interactive dashboard: `https://github.com/fernandogg2004/mlb-lineup-optimizer`. Software version `2.0.0@8ed6f63`.

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1 Introduction

Batting-order construction is among the oldest quantitative questions in baseball and a classical application of Markov chains and simulation in operations research [2, 3, 5, 8, 13, 15, 17]. The received wisdom from that literature, popularized by Tango et al. [18], is that the order matters, but only modestly: a sensible order is worth only a few runs per season less than the best one.

This paper re-derives that conclusion with three ingredients that we found missing from the software system it replaces:

1. **A precisely defined objective.** We maximize $\mathbb{E}[R_9 \mid \pi, \mathcal{I}]$, the expected runs in nine offensive innings for a fixed pregame order π given the information $\mathcal{I}$ available before first pitch. We do *not* optimize win probability, which at the level of a single game is close to a coin flip and depends on the opponent's offense as well.

2. **Exact evaluation and exhaustive search.** The predecessor system ranked orders by Monte Carlo estimates inside a genetic algorithm. Because selection among noisy estimates is biased upward (the "optimizer's curse" [16]), it could not certify that its output was optimal, and it overstated the gain of the selected order. We evaluate expected runs exactly (Section 4) and search all orders exhaustively (Section 5).

3. **Leakage-free, falsifiable validation.** All estimates for a game on date d use only plate appearances before d. We validate the plate-appearance model on held-out seasons, check that

the implied run expectancy matches observed scoring, and check the calibration of game-level run predictions (Sections 6 and 7).

What the paper does not claim. The value of an unplayed order cannot be observed: a historical game shows the runs scored under the order the manager chose, not under any other. Every comparison between orders in this paper is therefore a statement *about the model*. We validate the model's ingredients on observed data and quantify parameter uncertainty, but we make no causal claim that a team would have scored more runs (Remark 1).

Contribution. The mathematical tools are classical: absorbing Markov chains [12], the base-out run-expectancy framework [13, 18], and empirical-Bayes shrinkage [1, 6]. Our contribution is an end-to-end, reproducible implementation that (i) estimates the transition kernel and batter-specific outcome distributions without future information, (ii) evaluates lineups exactly with a proven error bound, (iii) certifies the optimum by exhaustive search, and (iv) reports uncertainty and validation honestly, including null and small effects. We do not claim mathematical novelty for the individual propositions; several are elementary consequences of standard theory, and we say so where relevant.

2 Problem formulation

2.1 Players, orders and feasibility

A lineup consists of nine hitters indexed by $i \in \{1, \dots, 9\}$. A batting order is a bijection $\pi : \{1, \dots, 9\} \to \{1, \dots, 9\}$ assigning hitter $\pi(t)$ to slot t; the set of all orders is the symmetric group $\mathcal{S}_9$, with $|\mathcal{S}_9| = 362{,}880$. Under the Official Baseball Rules any order of the nine starters is legal, and with the universal designated hitter (adopted by MLB in 2022) the pitcher does not bat, so the feasible set is $\mathcal{F} = \mathcal{S}_9$. Users of the dashboard may add constraints of the form "hitter i bats in slot t"; these define $\mathcal{F}_A = \{\pi : \pi(t) \in A_t \ \forall t\}$ for allowed sets A_t.

We consider a *fixed pregame* order. In-game substitutions, pinch hitting and defensive changes are outside the model: the nine starters bat throughout.

2.2 State space and outcomes

Definition 1 (Base-out states)**.** A transient state is a pair $s = (o, b)$ with $o \in \{0, 1, 2\}$ outs and $b \in \{0, 1\}^3$ the occupancy of first, second and third base. We encode $s = 8o + b_1 + 2b_2 + 4b_3 \in \{0, \dots, 23\}$ and add an absorbing state a (three outs). Let $\mathcal{S} = \{0, \dots, 23\}$ and $\bar{\mathcal{S}} = \mathcal{S} \cup \{\mathsf{a}\}$.

Definition 2 (Outcome classes)**.** The batter-specific result of a plate appearance (PA) is one of eight classes $\mathcal{C} = \{\mathrm{K}, \mathrm{BB}, 1\mathrm{B}, 2\mathrm{B}, 3\mathrm{B}, \mathrm{HR}, \mathrm{GB}, \mathrm{AIR}\}$: strikeout; walk, intentional walk, hit-by-pitch or catcher's interference; single; double; triple; home run; a ground ball not scored as a hit (including double plays, fielder's choices and reaching on an error); and a fly ball, line drive or pop-up not scored as a hit (including sacrifice flies).

Splitting non-hit balls in play by batted-ball type lets the model distinguish hitters who ground into double plays from those who hit sacrifice flies, without modelling either event per hitter.

Definition 3 (Transition kernel)**.** For $c \in \mathcal{C}$, $s \in \mathcal{S}$, $s' \in \bar{\mathcal{S}}$ and $r \in \{0, \dots, 4\}$, $J_c(s, s', r)$ is the probability that, from state s, a PA with outcome c (together with all base-running events up to the decisive pitch of the next PA) leads to s' with r runs scored. We write $P_c(s, s') = \sum_r J_c(s, s', r)$ and $\rho_c(s) = \sum_{s', r} r\, J_c(s, s', r)$.

Defining a transition from one decisive pitch to the next (Section 3.1) means stolen bases, pick-offs and wild pitches are part of the kernel, and an inning that ends on a caught stealing correctly leaves the batter at the plate to lead off the next inning.

2.3 Stochastic model

Fix an order π. Pitching regimes $g \in \{1, \dots, G\}$ index who pitches (here $G = 2$: the opposing starter and the opposing bullpen). Let $x_g(i) \in \Delta(\mathcal{C})$ be the outcome distribution of hitter i under regime g, and $w_k \in \Delta(\{1, \dots, G\})$ the regime distribution of inning $k \in \{1, \dots, 9\}$.

Assumption A1 (Markov plate appearances)**.** Within a half-inning pitched under regime g, given the current state $S_n = s$ and the batting slot $T_n = t$, the outcome $C_n \sim x_g(\pi(t))$ and the next state and runs $(S_{n+1},\, \Delta R_n) \sim J_{C_n}(s, \cdot, \cdot)$ are independent of everything else that happened before; then $T_{n+1} = T_n + 1 \pmod 9$.

Assumption A2 (Batter-independent kernel)**.** J_c does not depend on the identity of the batter or runners; base running is league-average.

Assumption A3 (Pitching changes between innings)**.** The regime G_k of inning k is drawn from w_k independently of the game state and of other innings, and does not change within an inning.

Assumption A4 (Outs)**.** Outs never decrease within an inning, and there is $q > 0$ such that every PA increases the number of outs with probability at least q whatever the state, hitter and regime: $\min_{g,i,s} \sum_c x_g(i)_c \sum_{s' :\, o(s') > o(s)} P_c(s, s') \geq q$, with $o(\mathsf{a}) = 3$.

Assumption A5 (Bookkeeping)**.** Every transition conserves participants: if $s' \neq \mathsf{a}$, $|b(s)| + 1 = |b(s')| + r + o(s') - o(s)$; if $s' = \mathsf{a}$, $r + 3 - o(s) \leq |b(s)| + 1$.

Assumptions A4 and A5 are checked on the estimated kernel: no observed transition decreases the outs, and every transition violating the identity is a data error and is removed (Section 3.1). Each inning starts in state $s = 0$ (no outs, bases empty); the extra-inning runner rule is irrelevant for a nine-inning objective.

Let $R_9 = \sum_{k=1}^{9} R^{(k)}$ be the runs scored in the first nine innings, where $R^{(k)}$ is the total of ΔR_n in inning k.

Definition 4 (Objective)**.** For an information set $\mathcal{I}$ (all data before the game date), the expected-runs objective is

$$V(\pi) = \mathbb{E}\big[R_9 \mid \pi, \mathcal{I}\big], \qquad \pi^\star \in \arg\max_{\pi \in \mathcal{F}} V(\pi),$$

where the expectation is with respect to the model of Assumptions A1 to A3 with parameters $\theta = (J, x, w)$ replaced by their estimates $\hat{\theta}$ from $\mathcal{I}$ (Section 3); $V = V_{\hat{\theta}}$ is a plug-in value, not a posterior-predictive expectation. Independence of the regimes across innings (Assumption A3) is a simplification (a starter who has left does not return), and w_k is interpretable as the probability that the starter pitches inning k only under it. Expected runs per inning ($V/9$) and per plate appearance are derived quantities; win probability is not modelled.

Remark 1 (Identification). For a historical game we observe one realization of R under the order π_{used}. $V(\pi)$ for $\pi \neq \pi_{\text{used}}$ is a counterfactual expectation that is identified only through the structural assumptions above. Assumptions A1–A3 are what make the order affect runs only through the sequence in which hitters come to the plate; they cannot be tested on games played under a single order. Moreover, the predicted gain of reordering (hundredths of a run) is two orders of magnitude smaller than the standard deviation of a team's runs in one game (about 3), so no feasible observational study could detect it reliably.

Table 1: Statcast data after processing (regular season and postseason). Non-conserving: bookkeeping violations outside censored transitions (data errors).

Season	Games	PAs	Reliable lineups	Censored	Non-conserving
2021	2,466	184,592	4,919/4,932	339	0
2022	2,470	185,047	4,923/4,940	222	1
2023	2,471	187,190	4,924/4,942	196	0
2024	2,472	185,715	4,928/4,944	215	0
2025	2,477	186,562	4,948/4,954	230	0
2026	2,469	186,777	4,924/4,938	205	1

3 Statistical estimation

3.1 Data

We use pitch-level Statcast data (Baseball Savant [14]) for 2021–2026, regular season and postseason, downloaded in two-day windows and stored unmodified, with SHA-256 manifests. For each PA the decisive pitch is the pitch with a non-empty `events` field; its fields `outs_when_up`, `on_1b/2b/3b` and `bat_score` give the state and score before the decisive pitch. The transition of PA n is from its decisive state to the decisive state of PA $n+1$ in the same half-inning, or to `a` for the last PA; its runs are the change in the batting team's score (for the last PA, up to the score at the end of the half-inning). Plate appearances interrupted by the end of an inning (`truncated_pa`) are not PAs. The final transition of a game's last half-inning is censored, and excluded from estimation, unless the game ended normally after three outs (a top half of the ninth or later with the home team ahead, or a bottom half with the visitors ahead); this covers walk-offs in any inning, called games and seven-inning games. Censoring is outcome-dependent (walk-off transitions score runs by construction), but it affects about one transition in a thousand.

Starting lineups are inferred as the first nine distinct batters of each team; the inference is marked reliable when the tenth PA returns to the leadoff hitter. In a random audit of 60 reliable 2026 team-games (E0) the inferred lineup and opposing starter matched the official boxscore in 60 of them. For the reference game we use the official boxscore (slot codes $100, \ldots, 900$). Table 1 summarizes the data.

3.2 Transition kernel

For a game on date d, let $m(d)$ be the first day of its month. The kernel is refitted monthly: $N_c(s, s', r)$ counts valid transitions in the window $[\max(m(d) - 1095 \text{ days}, \text{2023-01-01}), m(d))$ (if that window would span less than 180 days, the previous 365 days are used instead). Refitting monthly only discards information and cannot introduce look-ahead. The 2023 lower limit reflects the 2023 rule changes (larger bases, pick-off limits, defensive-shift restrictions), which changed base-running and batted-ball outcomes. The estimate is the Dirichlet posterior mean

$$\hat{J}_c(s, \cdot, \cdot) = \frac{N_c(s, \cdot, \cdot) + \alpha J_c^0(s, \cdot, \cdot)}{N_c(s) + \alpha}, \qquad \alpha = 2,$$

centred on the conventional deterministic advancement J^0 (runners advance one base on a single, a ground ball with a runner on first and fewer than two outs is a double play, other outs do not advance runners). The prior only matters in sparse cells.

3.3 Plate-appearance model

Let $h = (\text{batter side}, \text{pitcher hand})$ denote the handedness matchup and $\ell_h \in \Delta(\mathcal{C})$ the league outcome distribution for h. For hitter b facing pitcher p we model

$$\varphi(c \mid b, p) = \frac{\ell_h(c)\, m_b(c)\, m_p(c)}{\sum_{c'} \ell_h(c')\, m_b(c')\, m_p(c')}, \tag{1}$$

a multinomial generalization of the odds-ratio ("log5") method introduced by James [10] and discussed by Tango et al. [18], with batter and pitcher multipliers $m_b, m_p \in (0, \infty)^{\mathcal{C}}$. Platoon effects enter only through the league distribution ℓ_h; individual platoon splits are not estimated, because they stabilize only after very large samples [18].

Empirical-Bayes multipliers. Each PA in the 1,095 days before the cutoff receives a weight $\omega = 2^{-\text{age}/H}$, where the age is in days and H is a half-life. For hitter b let $n_b(c) = \sum \omega\, \mathbf{1}\{C = c\}$ and $e_b(c) = \sum \omega\, \ell_{h(\text{PA})}(c)$ over his PAs (the expected count under league rates). Treating $n_b(c)$ as Poisson with mean $m_b(c)e_b(c)$ and placing a Gamma(a_c, a_c) prior (mean one) on $m_b(c)$ gives the posterior

$$m_b(c) \mid \text{data} \sim \text{Gamma}\big(n_b(c) + a_c,\ e_b(c) + a_c\big), \qquad \hat{m}_b(c) = \frac{n_b(c) + a_c}{e_b(c) + a_c}. \tag{2}$$

Multipliers of hitters with little history therefore shrink to one, in the spirit of Efron and Morris [6] and Brown [1]. Pitchers are treated identically with strengths s_c. A team's bullpen is one pseudo-pitcher built from all PAs of pitchers other than that game's starter. The prior strengths are $a_c = \lambda_B\, \kappa_c$ and $s_c = \lambda_P\, \kappa_c$, with fixed relative strengths $\kappa = (12, 15, 40, 25, 6, 12, 60, 60)$ ordered by how fast each rate stabilizes. The scales (λ_B, λ_P) and H are tuned on 2024 only (Section 6).

Remark 2. With fractional weights, the Poisson model in (2) is a working likelihood, not a literal sampling model: decayed counts are not counts. We use (2) as a shrinkage rule whose prior strength is tuned out of sample, and the Gamma posterior as an approximate description of parameter uncertainty (Section 8).

Starter workload. For each starting pitcher, $w_k(\text{starter})$ is the decayed average, over his previous starts, of the share of inning-k plate appearances he faced, shrunk toward the league curve with a prior of 8 starts. The bullpen pitches the rest. The bullpen distribution for hitter b mixes (1) over pitcher hands in proportion to the bullpen's historical share of right-handed PAs.

3.4 Leakage control

Every quantity above, for a game on date d, uses only PAs with `game_date` $< d$ (same-day earlier games are excluded). This is enforced in code and tested: altering every PA on or after d leaves the fitted model bit-for-bit unchanged. Statcast columns that encode the future (e.g. `pitcher_days_until_next_game`) are never used.

4 Exact evaluation of a batting order

4.1 Half-innings as absorbing chains

Fix a regime g and order π. Within a half-inning the pair (S_n, T_n) is a Markov chain on $\bar{\mathcal{S}} \times \{1, \dots, 9\}$ by Assumption A1. Define the one-PA operator of hitter i,

$$M_g(i) = \sum_{c \in \mathcal{C}} x_g(i)_c \, P_c \in \mathbb{R}^{24 \times 25}, \qquad \rho_g(i) = \sum_c x_g(i)_c \, \rho_c \in \mathbb{R}^{24}.$$

Proposition 1 (Finite innings)**.** *Under Assumptions A1 and A4, the number N of PAs in a half-inning satisfies $\mathbb{P}(N > n) \le \mathbb{P}(\mathrm{Bin}(n, q) \le 2)$ for every n. In particular $N < \infty$ almost surely, $\mathbb{E}[N] \le 3/q$, and all moments of N are finite.*

Proof. Index PAs from 0 and let $U_n = \mathbf{1}\{o(S_{n+1}) > o(S_n)\}$ for the PA with index n. By Assumption A4, $\mathbb{P}(U_n = 1 \mid \mathcal{F}_n) \ge q$ on $\{N > n\}$, where $\mathcal{F}_n$ is the history before PA n. Outs never decrease and the inning ends once they reach three, so $N \le \min\{n : U_0 + \cdots + U_{n-1} \ge 3\}$. A standard coupling (draw $V_n \sim \mathrm{Unif}(0,1)$ independently and realize $U_n = \mathbf{1}\{V_n < p_n\}$ with $p_n = \mathbb{P}(U_n = 1 \mid \mathcal{F}_n) \ge q$; set $B_n = \mathbf{1}\{V_n < q\} \le U_n$) shows that N is stochastically dominated by the time T_3 of the third success in i.i.d. Bernoulli(q) trials B_n. Hence $\mathbb{P}(N > n) \le \mathbb{P}(T_3 > n) = \mathbb{P}(\mathrm{Bin}(n, q) \le 2)$ and $\mathbb{E}[N] \le \mathbb{E}[T_3] = 3/q$. Finite moments follow from the geometric tail of T_3. □

Lemma 2 (Runs are bounded by plate appearances)**.** *Under Assumption A5, every half-inning satisfies $R^{(k)} + \mathrm{LOB} + 3 = N$, where LOB is the number of runners left on base. Hence $R^{(k)} \le N - 3$ and $\mathbb{E}[R^{(k)}] \le 3/q - 3$.*

Proof. Summing the identity of Assumption A5 over the non-absorbing transitions of the inning and adding the inequality of the absorbing one gives $R^{(k)} + 3 \le N$. Defining $\mathrm{LOB} := |b(S_{N-1})| + 1 - r_N - (3 - o(S_{N-1})) \ge 0$ (the participants of the last transition that neither scored nor were put out) turns this into the identity $R^{(k)} + \mathrm{LOB} + 3 = N$. The bound on the expectation follows from Theorem 1. □

Proposition 3 (Inning tables)**.** *For a leadoff slot j and regime g, let $r_g(j) = \mathbb{E}[R^{(k)} \mid T_1 = j, G_k = g]$ and $L_g(j, j') = \mathbb{P}(\text{slot } j' \text{ leads off the next inning} \mid T_1 = j, G_k = g)$. Then, with $v_0 = \mathbf{1}_{\{0\}}^{\top}$ (all mass on state 0) and $v_{n+1} = v_n M_g(\pi(j+n))_{\cdot, \mathcal{S}}$, where slot indices are taken in $\{1, \dots, 9\}$ cyclically, $t \mapsto 1 + ((t-1) \bmod 9)$,*

$$r_g(j) = \sum_{n \ge 0} v_n \, \rho_g(\pi(j+n)), \qquad L_g(j, j') = \sum_{n \ge 0:\ j+n+1 \equiv j'} v_n M_g(\pi(j+n))_{\cdot, \mathsf{a}},$$

both series converging geometrically, and $L_g(j, \cdot)$ is a probability vector.

Proof. $v_n(s)$ is the probability that the inning is still in progress at PA n in state s (the slot is deterministic given n). The expected runs of PA n are $v_n \rho_g(\pi(j+n))$, and the inning ends at PA n with probability $v_n M_g(\cdot)_{\cdot, \mathsf{a}}$, in which case the next slot, $j + n + 1$, leads off the next inning. Summability and $\sum_{j'} L_g(j, j') = \mathbb{P}(N < \infty) = 1$ follow from Theorem 1; interchanging sum and expectation is justified by monotone convergence since runs are non-negative and $\mathbb{E}[R^{(k)}] < \infty$ (Theorem 2). □

Theorem 4 (Exact game value)**.** *Under Assumptions A1 to A5, let* $r_k = \sum_g w_{k,g} r_g$ *and* $L_k = \sum_g w_{k,g} L_g$*. The leadoff slots* $(J_k)_{k\geq 1}$ *of successive innings form a (time-inhomogeneous) Markov chain with* $J_1 = 1$ *and transition matrices* L_k*, and*

$$V(\pi) = \sum_{k=1}^{9} \mu_k^\top r_k, \qquad \mu_1 = e_1, \quad \mu_{k+1}^\top = \mu_k^\top L_k. \tag{3}$$

Proof. Every inning starts in state 0, so the only information an inning passes to the next is the slot of the next batter. By Assumptions A1 and A3, conditionally on $J_k = j$ the inning-k regime is independent of the past and, given the regime, the inning evolves as in Theorem 3; hence $\mathbb{P}(J_{k+1} = j' \mid J_1, \ldots, J_k = j) = \sum_g w_{k,g} L_g(j, j')$ and $\mathbb{E}[R^{(k)} \mid J_1, \ldots, J_k = j] = r_k(j)$. Taking expectations and summing over k gives (3). □

Theorem 4 is the batting-order analogue of the chain used by Bukiet et al. [2]; there, the full distribution of runs is propagated, while for the expected-value objective the nine-dimensional leadoff chain is sufficient. The run distribution, needed only for presentation, is obtained by propagating the joint law of (state, runs) within innings and of (leadoff, cumulative runs) across innings.

4.2 Truncation error

In practice each series in Theorem 3 is summed until the unabsorbed mass $\|v_n\|_1$ falls below a tolerance ε (we use 10^{-13}).

Proposition 5 (Truncation error)**.** *If every half-inning is truncated once its unabsorbed mass is at most* ε*, the computed value* $\hat{V}(\pi)$ *satisfies* $|V(\pi) - \hat{V}(\pi)| \leq 162\,\varepsilon/q$*.*

Proof. Fix an inning and let n be the truncation step, $\mathbb{P}(N > n) = \|v_n\|_1 \leq \varepsilon$. Runs after step n are at most the runners on base at step n (at most 3) plus the remaining PAs (each remaining participant scores at most once, Theorem 2). Given $\mathcal{F}_n$ on $\{N > n\}$, at most three more out increments are needed, so the expected number of remaining PAs is at most $3/q$ by the argument of Theorem 1. Hence $0 \leq r_g(j) - \hat{r}_g(j) \leq \varepsilon(3 + 3/q) \leq 6\varepsilon/q$. The computed $\hat{L}_g(j, \cdot)$ is substochastic with missing mass at most ε; writing $\delta_k = \mu_k - \hat{\mu}_k$, $\delta_{k+1}^\top = \delta_k^\top L_k + \hat{\mu}_k^\top (L_k - \hat{L}_k)$, so $\|\delta_{k+1}\|_1 \leq \|\delta_k\|_1 + \varepsilon$ and $\|\delta_k\|_1 \leq (k-1)\varepsilon$. Since $\|\hat{r}_k\|_\infty \leq \|r_k\|_\infty \leq 3/q$ by Theorem 2,

$$|V - \hat{V}| \leq \sum_{k=1}^{9} \Big(|\mu_k^\top (r_k - \hat{r}_k)| + |\delta_k^\top \hat{r}_k| \Big) \leq 9 \cdot \frac{6\varepsilon}{q} + \frac{3}{q} \sum_{k=1}^{9} (k-1)\varepsilon = \frac{162\,\varepsilon}{q}. \qquad \square$$

For the reference game the estimated floor is $q = 0.633$ and the largest truncated mass is below ε, so the bound is 1.7×10^{-11} runs, many orders of magnitude below the differences we report. The bound is tested numerically against coarse tolerances in the test suite.

4.3 Run expectancy and run values

Proposition 6. *For a single outcome distribution* x *(every hitter identical), let* $Q = \sum_c x_c P_c|_{\mathcal{S}\times\mathcal{S}}$ *and* $\rho = \sum_c x_c \rho_c$*. Under Assumption A4,* $I - Q$ *is invertible, the run-expectancy vector is* $\mathrm{RE} = (I-Q)^{-1}\rho$*, and the expected number of visits to each state starting from* $s = 0$ *is* $\nu^\top = e_0^\top (I-Q)^{-1}$*.*

Proof. The argument of Theorem 1 applies from any initial state, because the lower bound q of Assumption A4 is uniform in s; hence every row of Q^n tends to zero, the spectral radius of Q is below one and $(I - Q)^{-1} = \sum_{n\geq 0} Q^n$ (the fundamental matrix of the absorbing chain [12]). The identities follow by summing over PAs. □

The run value of class c is the visit-weighted mean change in run expectancy, $\mathrm{RV}(c) = \sum_s \nu(s)[\rho_c(s) + P_c(s,\cdot)\mathrm{RE} - \mathrm{RE}(s)]/\sum_s \nu(s)$ (with $\mathrm{RE}(\mathsf{a}) = 0$), i.e. linear weights [18]. We use these only to define the heuristic baselines and to describe hitters.

5 Optimization

5.1 Existence and exhaustive search

Proposition 7. *V attains its maximum on any non-empty $\mathcal{F}_A \subseteq \mathcal{S}_9$.*

Proof. $\mathcal{F}_A$ is finite and V is finite on it (Theorem 2). □

The question is computational. A direct evaluation of one order costs $O(9\,G\,\bar{n}\,|\mathcal{S}|^2)$ operations, where $\bar{n}$ is the number of PAs summed per half-inning (about 30–40 at $\varepsilon = 10^{-13}$). Exhaustive search exploits a symmetry.

Lemma 8 (Rotations)**.** *Let σ be the cyclic shift of slots, $(\pi \circ \sigma^m)(t) = \pi(t+m)$. The inning tables of $\pi \circ \sigma^m$ are those of π with indices shifted by m: $r_g^{\pi\sigma^m}(j) = r_g^{\pi}(j+m)$ and $L_g^{\pi\sigma^m}(j,j') = L_g^{\pi}(j+m, j'+m)$.*

Proof. A half-inning led off by slot j of $\pi \circ \sigma^m$ sees hitters $\pi(j+m), \pi(j+m+1), \ldots$, i.e. the same sequence as the inning led off by slot $j+m$ of π. □

Hence the 9! orders split into $8! = 40{,}320$ cyclic classes. For each class (fix hitter 1 in slot 1) we compute the tables once ($9G$ half-innings) and obtain the value of all nine rotations from (3) in $O(9 \cdot 9 \cdot 81\,G)$ further operations (nine rotations, nine innings). The complete search costs about $8! \cdot 9G\bar{n}|\mathcal{S}|^2 \approx 10^{10}$ floating-point operations, i.e. 1.40 seconds per instance on one laptop CPU with a compiled (Numba) implementation (measured while another experiment ran concurrently), and about ten seconds in a browser.

Corollary 9 (Certified optimum)**.** *Let $\hat{\pi}$ maximize the computed $\hat{V}$ and let $\Delta = \hat{V}(\hat{\pi}) - \max_{\pi \neq \hat{\pi}} \hat{V}(\pi)$. If $\Delta > 324\varepsilon/q$, then $\hat{\pi}$ is the unique maximizer of V; in any case $V(\hat{\pi}) \geq \max_\pi V(\pi) - 324\varepsilon/q$.*

Proof. Apply Theorem 5 to $\hat{\pi}$ and to a maximizer of V. □

For the reference game the computed margins are $\Delta = 8.2 \times 10^{-5}$ (Milwaukee) and 5.4×10^{-4} (Tampa Bay), many orders of magnitude above $324\varepsilon/q$ and above floating-point rounding (of order 10^{-14} in these sums).

5.2 Alternatives

We compare exhaustive search with (i) three heuristics without search: the guideline of Tango et al. [18] (best three hitters in slots 1, 2 and 4, next two in 3 and 5, the rest in decreasing run value), sorting by run value, and sorting by on-base probability; (ii) best-improvement 2-swap local search from the actual order, and from the actual order plus nine random starts; (iii) simulated annealing with swap moves; and (iv) a genetic algorithm with the operators and parameters of the predecessor system (ordered crossover, swap mutation, tournament selection, elitism), run both with the exact objective and, as in the original, with a 4,000-game Monte Carlo objective.

None of the heuristic or stochastic methods has an optimality guarantee. For the Monte Carlo variant, the reported value of the chosen order is biased upward. For any fixed set of orders with $\hat{V}(\pi) = V(\pi) + \eta_\pi$ and $\mathbb{E}\eta_\pi = 0$, $\mathbb{E}[\max_\pi \hat{V}(\pi)] \geq \max_\pi V(\pi)$ by Jensen's inequality; for an order

$\hat{\pi}$ selected on such estimates, the post-decision surprise $\mathbb{E}[\hat{V}(\hat{\pi}) - V(\hat{\pi})]$ is positive [16], and this is what E4a measures. Exact evaluation removes the Monte Carlo component of this bias. It does not remove the selection effect of estimation error in the inputs: $\hat{\pi}$ maximizes the plug-in value $V_{\hat{\theta}}$, whose maximum overstates the expected gain under parameter uncertainty (Section 7.6).

6 Experimental methodology

The protocol (`docs/methodology/experimental-protocol.md`) was fixed before any 2025–2026 metric was computed. Hyperparameters were tuned on 2024 only; 2025 and 2026, including the reference game, are test seasons.

E1: kernel validation. For each season 2023–2026 we compare model run expectancy (Theorem 6, with that season's kernel and league outcome mix) with empirical run expectancy: the mean runs scored from each state to the end of the half-inning, excluding walk-off half-innings.

E2: plate-appearance models. Rolling origin with 14-day blocks: the model is refitted at the start of each block on data before it and predicts every PA in the block. Candidates: M0 (league distribution by handedness), M1 (+ batter multipliers), M2 (+ pitcher multipliers; the selected model), M2 with untuned priors, and M3, a gradient-boosted multinomial classifier (LightGBM [11]) on the M2 inputs, trained on rolling 2023–2024 snapshots. The metrics are log-loss and Brier score (strictly proper [9]) and class-wise calibration, with 95% confidence intervals for differences from a game-level cluster bootstrap [7].

E3: game-level calibration. For every 2025–2026 team-game with a reliable lineup we compare the predicted $\mathbb{E}[R_{1..8}]$ with the observed runs in innings 1–8 (the home team does not always bat in the ninth). We report bias, the calibration slope of observed on predicted, mean squared error relative to a constant (previous-season mean) and to the team's mean over its previous 60 games, and randomized PIT values [4] of the observed runs under the exact predicted distribution for a random subset.

E4: optimization. (a) Search methods on 200 random 2026 regular-season team-games: gap to the exhaustive optimum, share of instances solved exactly, evaluations, runtime; the Monte Carlo genetic algorithm on the first 30 of them, where the order used is also evaluated so that the two are compared on the same instances. (b) For all 2025–2026 team-games: the model gain of the optimum over the order used, and the percentile of the order used among all 9! orders. (c) Parameter uncertainty on 80 random 2026 team-games: 40 joint posterior draws (2) of all batter, starter and bullpen multipliers per game; for each draw we re-evaluate the point-estimate optimum and the order used, and re-optimize.

E5: reference game. 2026 World Series Game 5 (gamePk 849799), both teams, with 400 posterior draws and sensitivity to the kernel era and to untuned priors.

All computations are deterministic given seeds; the Statcast extraction, model, evaluator and experiments are in the repository, and every number in this paper is read from the `outputs/` files produced by `batting-order experiments`.

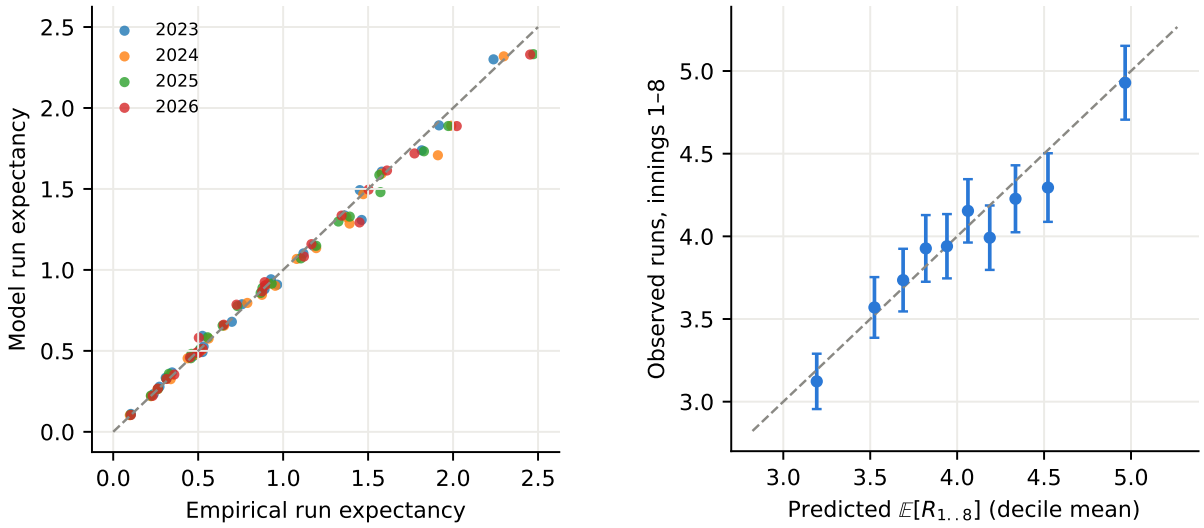


Figure 1: Left: model versus empirical run expectancy for the 24 base-out states, 2023–2026 (E1). Right: observed runs in innings 1–8 versus predicted expected runs, by decile of the prediction, 2025–2026 test seasons, with ±2 standard errors (E3).

7 Results

7.1 Data and kernel (E1)

Table 1 summarizes the processed data. Outside the censored end-of-game transitions, bookkeeping violations (scorer errors) number between 0 and 10 per season. The kernel reproduces observed run expectancy at the start of an inning (state 0) within 0.007 runs in every season 2023–2026 (e.g. RE(0) = 0.491 in 2026), and the visit-weighted mean absolute difference over all 24 states is at most 0.011 (Fig. 1). The largest differences (up to 0.20 runs) occur in crowded no-out states (for example first and third, or bases loaded), where the model is *below* the empirical value. Two observations suggest that this is mainly a selection effect rather than a kernel error: the expected immediate runs of each outcome out of first-and-third with no outs match the data within 0.014 runs (2024), and the outcome mix observed in that state is markedly more favourable to the offence than the league mix (18.7% singles and 19.9% strikeouts versus 14.1% and 22.6% overall). Innings reach those states disproportionately against weak pitching or with strong hitters at the plate, which a league-average run-expectancy calculation cannot reflect. Censoring of end-of-game transitions, which score runs by construction, pushes in the same direction but is too rare to explain the gaps. Game-level predictions (E3) use the actual hitters and pitchers and are the relevant check.

7.2 Plate-appearance model (E2)

Tuning on 2024 selected prior-strength scales $\lambda_B = 0.50$, $\lambda_P = 0.75$ and half-life $H = 365$ days (half-lives searched: 180, 270, 365, 730, 1095 days); all three selected values are interior to the final grid. Table 2 reports the test seasons. Batter multipliers reduce log-loss relative to the league model by 0.017 nats per PA (mean of the two seasons), and pitcher multipliers by a further 0.010; both differences are large relative to their bootstrap intervals. Tuning the priors matters little (0.0010). The gradient-boosted model M3, using the same inputs, differs from M2 by −0.0011 nats in 2025 (95% interval −0.0015 to −0.0007) and −0.0002 in 2026 (−0.0006 to +0.0001). Class-wise

Table 2: Held-out log-loss (nats per PA) and difference from M2 with 95% game-cluster bootstrap intervals; negative differences favour the row model. 2025: 186,562 PAs; 2026: 186,777 PAs.

	2025		2026	
Model	Log-loss	Δ vs M2	Log-loss	Δ vs M2
M0: league, by handedness	1.7724	+0.0278 [+0.0267, +0.0289]	1.7735	+0.0265 [+0.0255, +0.0276]
M1: + batter	1.7547	+0.0102 [+0.0095, +0.0109]	1.7563	+0.0093 [+0.0087, +0.0100]
M2: + pitcher (selected)	1.7446	—	1.7470	—
M2 with untuned priors	1.7457	+0.0011 [+0.0009, +0.0013]	1.7478	+0.0009 [+0.0007, +0.0010]
M3: gradient boosting on M2 inputs	1.7435	−0.0011 [−0.0015, −0.0007]	1.7467	−0.0002 [−0.0006, +0.0001]

calibration of M2 is shown in Fig. 2.

Deviation from the pre-specified selection rule. Our protocol selected the simplest model not significantly worse than the best; strictly applied to 2025, it selects M3. We nevertheless use M2 for the lineup analysis and report this as a deviation. The reasons are that M3's advantage is about one thousandth of a nat per PA in one of the two test seasons and absent in the other, and that M2 provides the posterior distribution used for parameter uncertainty (Section 7.6), which M3 does not. We checked what the choice changes on 300 random test team-games (E2c). The *magnitude* of the conclusions does not depend on it: the mean gain of the optimal order over the actual order is 0.0197 runs per game under M2 and 0.0196 under M3. The *identity* of the optimal order does: the two models select the same exact order in only 3.0% of the games, and the M2 optimum is worse under M3 than the M3 optimum by 0.0073 runs per game on average (maximum 0.0379); the correlation of the two models' values over all 9! orders averages 0.841. This is the same phenomenon as in Section 7.6: many orders are nearly tied, and small changes in the inputs change which one is best. One caveat applies to M3: for the bullpen pseudo-pitcher we supplied a sample-size feature outside its training range, so M3's absolute run levels for innings against the bullpen should not be relied on.

The original model. On the window where its stored features exist (2026-03-20 to 2026-06-12, 77,651 PAs, which are out of sample for it), the predecessor system's 51-feature LightGBM has log-loss 1.4383 on the common seven-class taxonomy, against 1.4360 for M2 and 1.4554 for the league model: M2 is better by 0.0024 nats per PA (95% interval 0.0003 to 0.0050), despite the legacy features being recomputed daily rather than every two weeks.

7.3 Run predictions at the game level (E3)

Over 9,860 test team-games, the mean prediction error (observed minus predicted runs in innings 1–8) is −0.035 runs (95% interval −0.096 to +0.026), and the calibration slope is 0.94 (95% interval 0.81 to 1.06); neither indicates miscalibration (Fig. 1). The mean squared error is 9.459, compared with 9.680 for the previous season's league mean and 9.781 for the team's mean over its previous 60 games. The correlation between predicted and observed runs is only 0.151: almost all of the variance of a single game's runs is unpredictable before the game. Randomized PIT values of the exact predicted run distributions on 300 random team-games are close to uniform, and the central 80% predictive interval covers 82.7% of outcomes (Fig. 3). Because the outcome is an integer, this "central 80%" interval holds on average 85.5% of the model's own probability, and observed coverage should be compared with that value: the difference is −1.4 binomial standard errors,

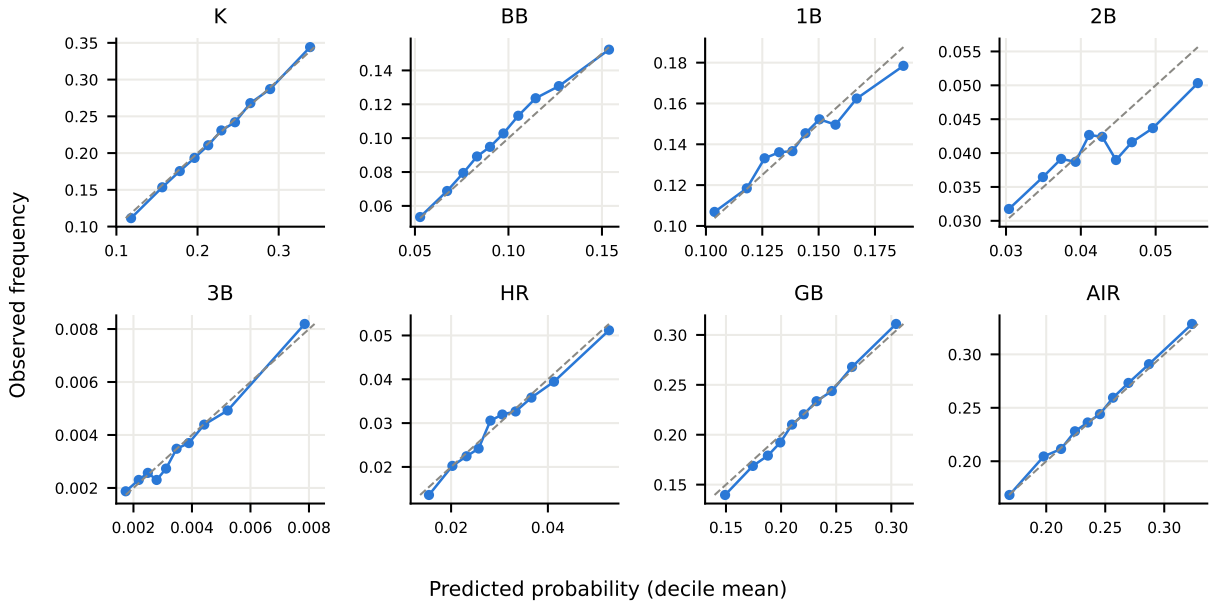


Figure 2: Class-wise calibration of M2 on 2026 (deciles of the predicted probability).

consistent with calibration of the predictive distribution. Team-games that batted fewer than eight innings (12, e.g. called games) are excluded from E3.

7.4 Search methods (E4a)

Table 3 compares the search methods with the exhaustive optimum. Exhaustive search takes 1.40 s per instance on one machine (all 362,880 orders). Local search from ten random starts found the optimum in every instance tested, and simulated annealing and the genetic algorithm with exact fitness came close, so for this problem size the choice among competent search methods is unimportant once the objective is exact; exhaustive search is preferred because it provides the certificate of Theorem 9. The heuristic guideline of Tango et al. [18] recovers about half of the gap between the actual order and the optimum.

The genetic algorithm with Monte Carlo fitness, as in the predecessor system, ends about as far from the optimum as the managers' own orders: on the same 30 instances its mean gap is 0.0223 runs, against 0.0240 for the order actually used. Its reported value of the chosen order, however, exceeds that order's exact value by 0.164 runs on average: the optimizer's curse is an order of magnitude larger than the effect being optimized. (The heuristic orders in Table 3 use run values averaged over the starter and bullpen regimes; in E4b they use the starter regime only, which explains the small difference between the two experiments' heuristic gaps.) With 4,000 simulated games, the standard error of one fitness evaluation is about $3/\sqrt{4000} \approx 0.05$ runs, larger than the differences among good orders.

7.5 How far are real orders from the optimum? (E4b)

Across 9,860 team-games of 2025–2026, the model-optimal order exceeds the order actually used by 0.020 expected runs per game on average (median 0.018, 90th percentile 0.033, maximum 0.080), or 3.2 runs per 162 games if the gap persisted every day (Fig. 4). The best and the worst of all

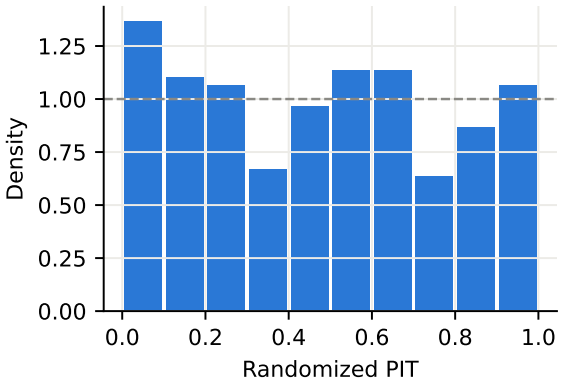


Figure 3: Randomized PIT histogram of observed runs in innings 1–8 (E3).

Table 3: Search methods on 200 random 2026 regular-season team-games. Gap: exact $V(\pi^\star) - V(\hat{\pi})$ in runs per game; Optimal: share of instances where the exact optimum was returned. †On 30 of them.

Method	Mean gap	Max gap	Optimal	Evaluations
Exhaustive search (exact)	0.0000	0.0000	100%	362,880
2-swap local search, 10 starts	0.0000	0.0000	100%	1,933
Simulated annealing	0.0000	0.0021	94%	1,707
Genetic algorithm, exact fitness	0.0001	0.0034	82%	523
2-swap local search from actual order	0.0004	0.0091	68%	203
Tango et al. guideline	0.0092	0.0251	0%	0
Sort by run value	0.0092	0.0306	1%	0
Sort by on-base probability	0.0110	0.0341	0%	0
Order actually used	0.0186	0.0641	0%	0
Original GA, Monte Carlo fitness†	0.0223	0.0526	0%	1,710

9! orders differ by only 0.094 runs per game on average, and the optimum exceeds the average of a uniformly random order by 0.050. Orders actually used rank, on median, above 97.8% of all possible orders, and 82% of them are above the 90th percentile. In this model, therefore, managers' orders are close to optimal and the room for improvement is a few runs per season, in line with the classical literature [8, 18].

7.6 Parameter uncertainty

In 80 random 2026 team-games with 40 posterior draws each (E4c), the point- estimate optimal order is better than the actual order in 87.3% of draws on average, but the 90% posterior interval of the gain excludes zero in only 33.8% of games, and its mean width (0.051 runs) exceeds the typical gain. The exact optimal order changes in almost every draw (it is unchanged in 0.5% of draws): many orders are nearly tied, and the data cannot single out one of them. What the data do support is a statement about a set of good orders (those that move the best hitters up), not about a unique best order.

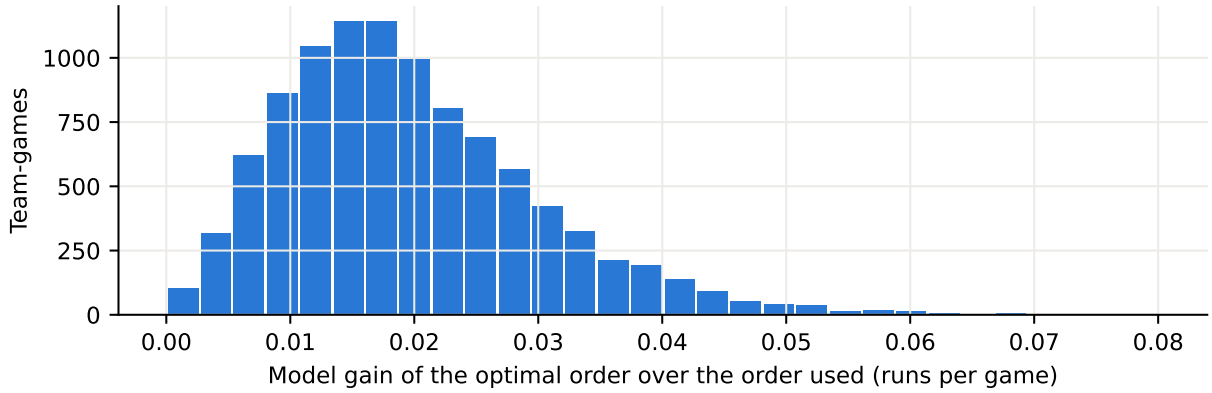


Figure 4: Model gain of the optimal order over the order actually used, 2025–2026 (E4b).

Table 4: 2026 World Series Game 5: starting orders used and model-optimal orders (information before 2026-10-28).

	Milwaukee Brewers		Tampa Bay Rays	
Slot	Used	Optimal	Used	Optimal
1	Jackson Chourio	Brice Turang	Yandy Díaz	Liam Hicks
2	Jake Bauers	Jake Bauers	Jonathan Aranda	Jonathan Aranda
3	Christian Yelich	William Contreras	Junior Caminero	Junior Caminero
4	William Contreras	Jackson Chourio	Liam Hicks	Yandy Díaz
5	Brice Turang	Cooper Pratt	Jonny DeLuca	Chandler Simpson
6	Sal Frelick	Christian Yelich	Chandler Simpson	Ryan Vilade
7	Cooper Pratt	Garrett Mitchell	Ben Williamson	Jonny DeLuca
8	Garrett Mitchell	Sal Frelick	Ryan Vilade	Ben Williamson
9	David Hamilton	David Hamilton	Taylor Walls	Taylor Walls

7.7 The 2026 World Series, Game 5 (E5)

Milwaukee won 4–1 at Tropicana Field. Facing Drew Rasmussen and the Tampa Bay bullpen, the order Milwaukee used has model value 3.833 expected runs over nine innings; the optimal order 3.843 (gain +0.010, 90% interval −0.002 to +0.021, optimal better in 91.2% of 400 posterior draws; the exact top order recurs in 0.2% of draws, and its mean regret against each draw's own optimum is 0.013 runs, as large as the point gain). The order used ranks above 97.9% of all orders. For Tampa Bay, facing Jacob Misiorowski and the Milwaukee bullpen, the values are 3.336 and 3.360 (gain +0.024, interval +0.009 to +0.041, better in 99.5% of draws; rank 99.3%). The observed scores (4 and 1 runs) are single draws from predictive distributions with a standard deviation of about three runs and carry essentially no information about the merit of either order (Remark 1). Table 4 lists the orders.

8 Sensitivity, limitations and threats to validity

Parameter uncertainty. The Gamma posteriors describe sampling uncertainty in player multipliers under a working Poisson likelihood; they do not include uncertainty in the kernel, the

starter-workload curve, or model misspecification, and draws are independent across classes and players. The weighted working likelihood, on the other hand, overstates the sampling variance of decayed counts, and kernel uncertainty is shared by all orders and partly cancels in differences. The reported intervals are therefore likely, but not guaranteed, to understate total uncertainty.

Misspecification. Assumption A2 ignores runner speed and individual double-play tendencies beyond ground-ball rate; fast runners make the leadoff slot more valuable than the model suggests. Assumption A3 places pitching changes between innings, whereas real changes often happen mid-inning and respond to the score (and, through the times-through-the-order effect, to the lineup). Platoon effects are league-level. In-game substitutions are ignored, which matters most for slots 7–9 in the late innings.

Objective. Maximizing expected runs is not the same as maximizing win probability; in close late-game situations a manager may value the variance of runs differently. We do not model this.

Data. Statcast occasionally revises recent games; decisive-pitch states may contain scorer errors (removed when they violate bookkeeping: 0–10 transitions per season outside the censored ones, Table 1). Lineups for the backtests are inferred (60 of 60 matched the official boxscore in the E0 audit), with no guarantee for every game.

Counterfactuals. As stated in Remark 1, none of the comparisons between orders is an observed effect.

9 Conclusions

Under explicit Markov assumptions, the expected runs of a batting order can be computed exactly and cheaply, which turns batting-order optimization from a noisy search into an exhaustive one with a certificate. With an empirical-Bayes plate-appearance model that uses only pregame information, the model's run predictions are calibrated on two held-out seasons, and the simple model is at least as good as the predecessor system's 51-feature gradient-boosting model.

The substantive conclusion is modest and, we think, robust: in this model the batting orders that MLB teams actually used in 2025–2026 were close to optimal (median rank above 97.8% of all orders), and reordering would add about 0.020 expected runs per game, roughly 3.2 runs over a season. On average, the best and worst possible orders of the same nine hitters differ by less than a tenth of a run per game (0.094). For the 2026 World Series Game 5 the estimated gains were +0.010 (Milwaukee) and +0.024 (Tampa Bay) runs, with parameter uncertainty of the same size, and the identity of the single best order is not statistically determined. We find no evidence that either manager's order was materially worse than the model's recommendation, and the actual result of the game cannot speak to that question.

What remains uncertain is everything outside the model: base-running speed, mid-inning pitching changes, individual platoon effects and in-game substitutions. Each of these could change the ranking of nearly tied orders; we expect none of them to change the order of magnitude of the effect. The method itself is the paper's practical contribution: a reproducible, leakage-free pipeline and an exact optimizer, with an interactive public interface that runs the same computation in the browser.

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A Reproducibility

```
pip install -e ".[ingest,compare,figures,dev]"
batting-order ingest 2021 2022 2023 2024 2025 2026   # Baseball Savant, resumable
batting-order build  2021 2022 2023 2024 2025 2026
batting-order experiments                             # writes outputs/*.json
python scripts/make_paper_assets.py                   # figures + generated/*.tex
cd research && latexmk -pdf paper.tex
```

Raw files are verified against `data/raw/statcast/season=*/MANIFEST.json`. The numerical claims of Section 4 are covered by tests: closed-form checks (an out/home-run model where runs per inning are negative binomial with mean $3(1-q)/q$), agreement with an independently written Monte Carlo simulator, agreement of the exact distribution's mean with (3), the truncation bound of Theorem 5, and parity of the browser implementation with the reference implementation to 10^{-10}.

B Run expectancy, 2026

Model run expectancy (empirical in parentheses) by bases occupied and outs; 2026 kernel with the league outcome mix (E1). Bases: 1 = first, 2 = second, 3 = third.

Bases	0 outs	1 out	2 outs
—	0.491 (0.493)	0.267 (0.263)	0.104 (0.104)
1–	0.868 (0.877)	0.513 (0.526)	0.222 (0.232)
-2-	1.082 (1.122)	0.661 (0.650)	0.327 (0.314)
12-	1.496 (1.503)	0.924 (0.891)	0.459 (0.451)
–3	1.293 (1.450)	0.904 (0.893)	0.354 (0.359)
1-3	1.719 (1.773)	1.159 (1.167)	0.489 (0.504)
-23	1.888 (2.022)	1.335 (1.344)	0.581 (0.504)
123	2.330 (2.454)	1.614 (1.611)	0.785 (0.726)